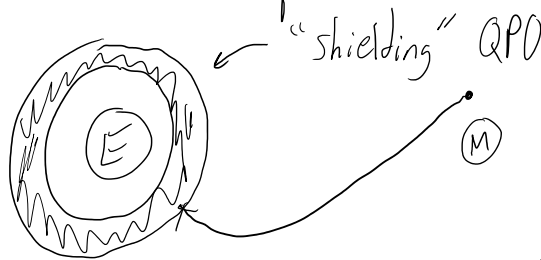


• Motivation:

- Dynamical systems in celestial mechanics often exhibit chaos (traj. are sensitive to initial perturbations)
- Regular solutions include traj. with repeated behaviour as $t \rightarrow \infty$ (eq'm pt, POs, QPOs)
- Regular solutions provide a skeleton for allowable motion in chaotic systems (e.g. QPOs block chaotic transport between the Earth & the Moon)



- How do we find these regular sol'ns in a Hamiltonian system?

Checkout: ASEN 6060, APPM 5460

- Goal: Develop a procedure to identify & compute regular sol'ns to understand allowable motion

Setup: Let $\vec{q}$ denote our configuration space (position-space) coordinates & let $\vec{p}$ be our conjugate momenta for an n DOF system. Let $\vec{z} = \begin{bmatrix} \vec{q} \\ \vec{p} \end{bmatrix}$ be our $2n$ -dimensional particle state. For a Hamiltonian system, motion is given by,

$$\dot{\vec{z}} = J \frac{\partial H}{\partial \vec{z}}$$

where $J = \begin{bmatrix} 0 & -I \\ I & 0 \end{bmatrix}$ & H is the system Hamiltonian (1HR4BP)

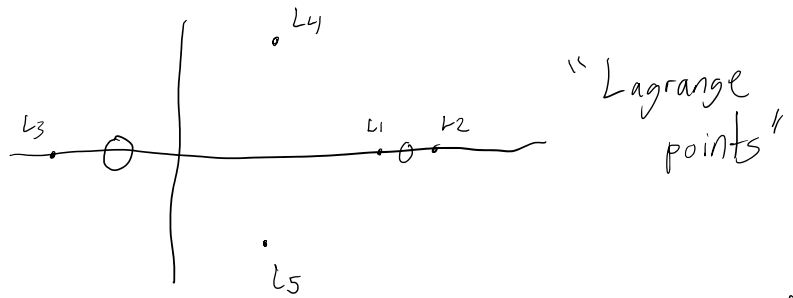
- We call a system autonomous if $H \equiv H(\vec{z}; \vec{\mu})$ and non-autonomous if $H \equiv H(\vec{z}, t; \vec{\mu})$, where $\vec{\mu}$ is a vector of dynamical parameters (12BP, CR3BP)

① Identify "simplest" solutions

- In autonomous systems, we solve for eq'm from $\dot{\vec{z}} = J \frac{\partial H}{\partial \vec{z}} = \vec{0}$. In

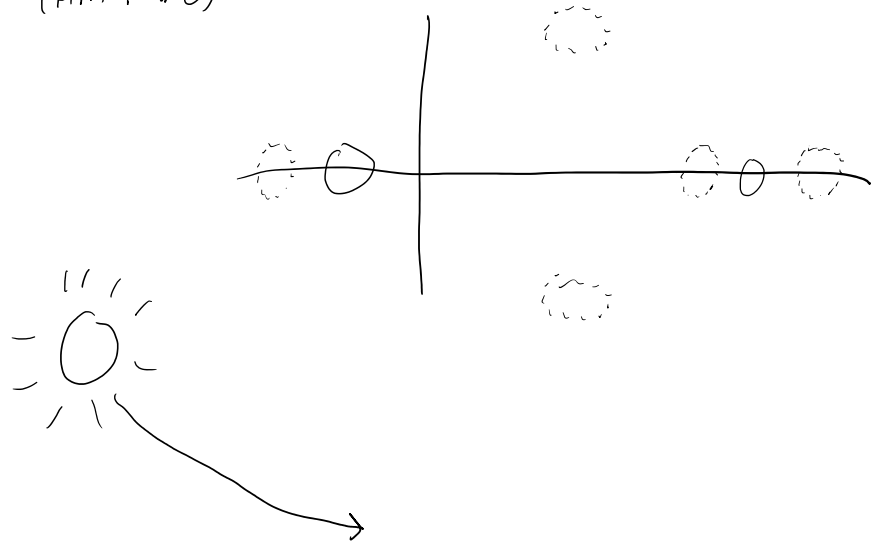
Hamiltonian systems, these are isolated (no two eq'm are touching)

Example: CR3BP
(rotating frame)



→ In non-autonomous systems, the existence of regular sol'ns depends on the explicit time dependence in $H(\vec{x}, t; \vec{p})$. If time dependence is periodic or quasi-periodic, simplest sol'ns are POs or QPOs

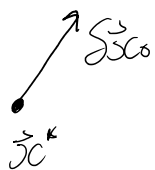
Example: Hill Restricted Four-Body Problem (HR4BP)
(Hill Frame)



② Analyze the neighbourhood of space surrounding "simplest" sol'ns

Autonomous Systems

→ Last time: Linear analysis about eq'm $\vec{x}^*$



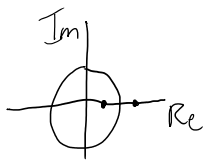
Eigenvalues of the A matrix determine local motion

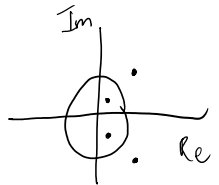
$$\dot{\vec{S}}_{\vec{x}} = J \left. \frac{\partial^2 H}{\partial \vec{x}^2} \right|_{\vec{x}^*} \vec{S}_{\vec{x}} = A \vec{S}_{\vec{x}} \rightarrow \vec{S}_{\vec{x}}(t) = \int \vec{S}_{\vec{x}c0} e^{At} dt = e^{At} \vec{S}_{\vec{x}c0}$$

→ Often, the eigenvectors of A matrix are linearly independent and span

the phase space about $\vec{z}^*$

→ Let $\vec{u}_j \in \mathbb{C}^{2n}$ be an eigenvector of A with eigenvalue $\sigma_j \in \mathbb{C}$

 → If $\text{Im}(\sigma_j) = 0$, let $\vec{u}_j = \vec{r}_j$ & $\sigma_j = \gamma_j$ (real evals & evs)
 $\vec{r}_j \in \mathbb{R}^{2n}, \gamma_j \in \mathbb{R}$

 → If $\text{Im}(\sigma_j) \neq 0$, let $\vec{u}_j = \vec{s}_j \pm i\vec{w}_j$ & $\sigma_j = \alpha_j \pm i\beta_j$
 where $\vec{s}_j, \vec{w}_j \in \mathbb{R}^{2n}, \alpha_j, \beta_j \in \mathbb{R}$

→ When the eigenvectors are lin. ind., we can always write:

$$\vec{S}(t) = e^{At} \vec{S}_0 = \sum_{\text{real}} c_j^{\pm} e^{\pm \gamma_j t} \vec{r}_j^{\pm} + \sum_{\text{complex}} e^{\pm \alpha_j t} \begin{bmatrix} \vec{s}_j & \vec{w}_j \end{bmatrix} \begin{pmatrix} \cos \beta_j t & \sin \beta_j t \\ -\sin \beta_j t & \cos \beta_j t \end{pmatrix} \begin{bmatrix} c_j' \\ c_j'' \end{bmatrix}$$

→ What geometric insight do we get from this expression?

Linearly independent evs $\Rightarrow \{\vec{r}_j, \vec{s}_j, \vec{w}_j\}$ form a basis for the phase space about $\vec{z}^*$

Evaluating $\vec{S}(t)$ at $t_0 = 0$, we have,

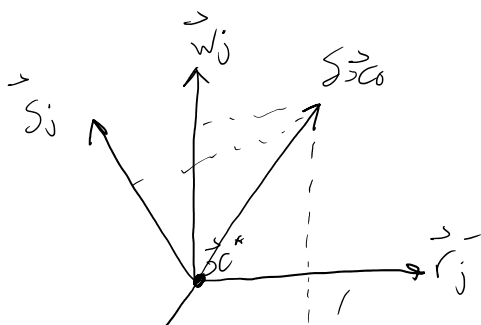
$$\vec{S}(t_0) = \vec{S}_0 = \sum_{\text{real}} c_j^{\pm} \vec{r}_j^{\pm} + \sum_{\text{complex}} \begin{bmatrix} \vec{s}_j & \vec{w}_j \end{bmatrix} \begin{bmatrix} c_j' \\ c_j'' \end{bmatrix}$$

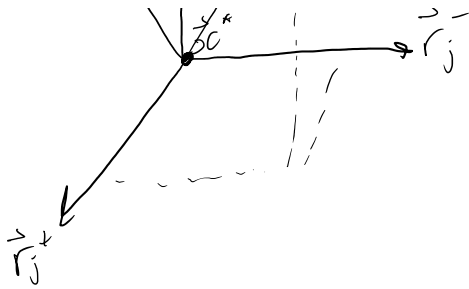
This is a decomposition of $\vec{S}_0$ along the basis vectors $\{\vec{r}_j, \vec{s}_j, \vec{w}_j\}$

Two Important Cases:

(A) "Real" Case:

$\vec{S}_0$ only has components along a single $\vec{r}_j^{\pm}$ pair

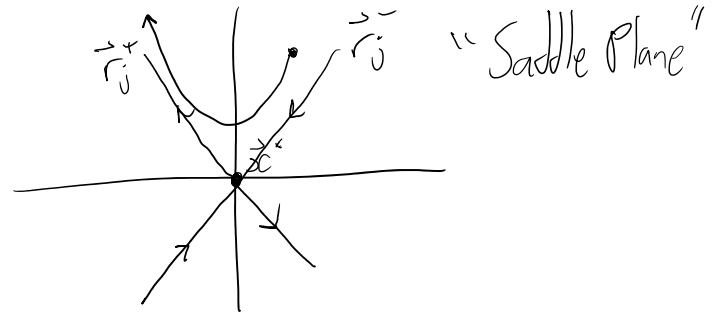




also only two components along a single pair

Our general sol'n $\vec{S}(t)$ reduces to:

$$\vec{S}(t) = c_j^+ e^{\gamma_j t} \vec{r}_j^+ + c_j^- e^{-\gamma_j t} \vec{r}_j^-$$



(B) "Imaginary" Case

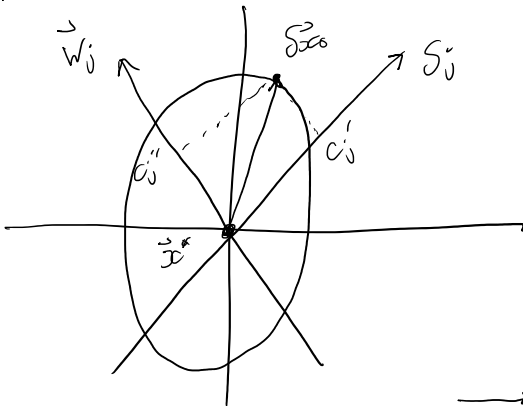
$\vec{S}_{co}$ only has components along a single $\vec{S}_j, \vec{w}_j$ pair, where $\alpha_j = 0$ ($\sigma_j = \pm i\beta_j$)

Our general sol'n $\vec{S}(t)$ reduces to:

$$\vec{S}(t) = \begin{bmatrix} \vec{S}_j & \vec{w}_j \end{bmatrix} \begin{pmatrix} \cos\beta_j t & \sin\beta_j t \\ -\sin\beta_j t & \cos\beta_j t \end{pmatrix} \begin{bmatrix} c_j' \\ c_j'' \end{bmatrix}$$

2×1

"Center Plane"

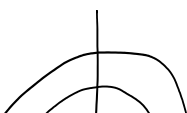


$\Rightarrow$ Motion in the center plane is locally periodic in the vicinity of the eq'm

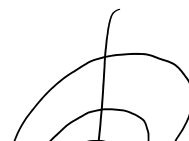
$\hookrightarrow$ Periodic orbits surround equilibria that have purely imaginary evals

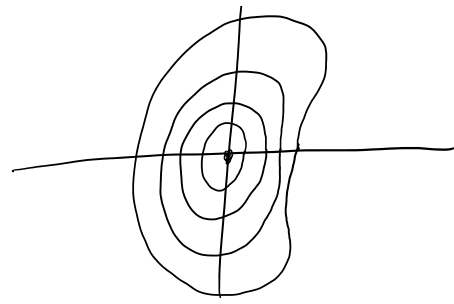
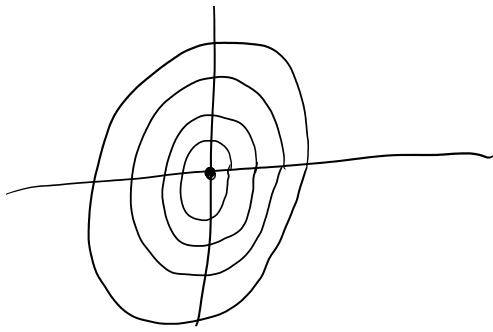
$\rightarrow$ By varying $\vec{S}_{co}$ in the center plane, we get a family of periodic orbits about the eq'm (Lyapunov Center Theorem)

Linear Analysis



True, Nonlinear System





→ Motion is no longer accurately characterized by linear analysis "far away" from the eq'm $\vec{x}^*$. Yet, POs often exist far from equilibrium. How do we find & compute these POs?

③ Compute & Continue POs

→ POs far from $\vec{x}^*$ are still part of the same family as the local POs about $\vec{x}^*$ (local Center Manifold Theorem)

→ A periodic orbit is a sol'n to $\dot{\vec{x}} = \mathcal{J} \frac{\partial H}{\partial \vec{x}}$ such that,

$$\vec{x}(kT + t_0) = \vec{x}(t_0)$$

where T is the orbit period & $k=1,2,3, \dots$

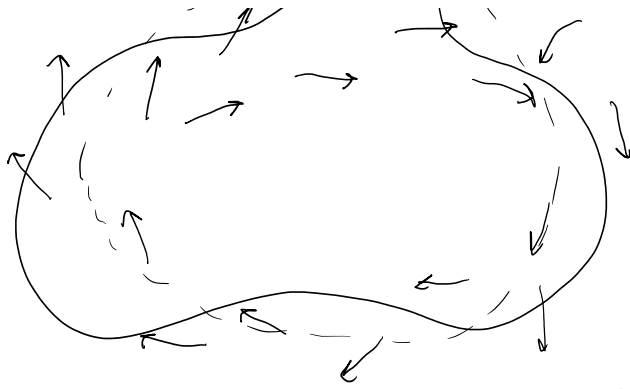
We want to "compute" a PO in a Hamiltonian system, meaning we can obtain a set of states $\vec{x}_j$ on the PO such that:

$$\Phi_T(\vec{x}_j) = \vec{x}_j$$

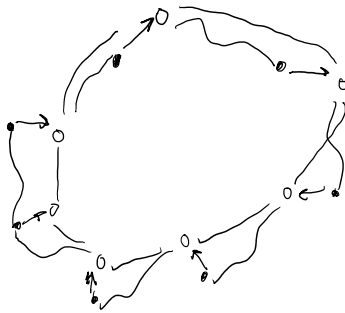
where $\vec{x}(t) = \Phi_t(\vec{x}_0)$ is the solution flow.

i) Collocation techniques: Find a polynomial describing the PO that also aligns with the vector field $\mathcal{J} \frac{\partial H}{\partial \vec{x}}$





ii) Shooting techniques : Correct $\vec{x}_0$ with dynamics until constraints are satisfied



Single-Shooting

→ We are interested in correcting a single $\vec{x}_0$ until it falls onto a periodic orbit

→ Works best for "smaller" periodic orbits

→ Introduce the periodicity constraint:

$$\Phi_T(\vec{x}_0) = \vec{x}_0 \Rightarrow \Phi_T(\vec{x}_0) - \vec{x}_0 = \vec{0}$$

→ Goal: Find $\vec{x}_0$ & T which satisfy this constraint

Our problem boils to a root-finding problem!

→ Let $\vec{z} = \begin{bmatrix} \vec{x}_0 \\ T \end{bmatrix}$ denote our free variable vector & let

$f(\vec{z}) = \Phi_T(\vec{x}_0) - \vec{x}_0$, then we need to solve:

$$f(\vec{r}) = \vec{0}$$